

 Announcement

Kindly note: We're currently experiencing a high volume of inquiries and calls, potentially leading to delays in our initial response.



Dermatology 23S Module 2

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Go to Week 1 Academic Forum

Scenario 3

 Settings ▾[◀ Scenario 2](#) Display replies in nested form**Scenario 3**

by [Derek Kyanja \(Tutor\)](#) - Tuesday, 7 November 2023, 7:28 PM

Hi Group,

Here is scenario 3.

A 38-year-old man presents with a well-demarcated, skin-coloured to red dermal nodule on his forehead, which has a punctum. Some cyst contents are expressed with pressure and it has an odour.

Please answer all the questions below (please read the carefully).

1. What is the most likely diagnosis?
2. What condition is associated with these types of skin lumps?
3. A 56-year-old woman with a BMI of 30 kg/m² and depression, presents with multiple painful lipomas. Which condition is this most likely to be associated with?

89 words

[Reply](#)**Re: Scenario 3**

by [Sophie Holloran](#) - Tuesday, 7 November 2023, 9:10 PM

1. Without an image, I would say the most likely diagnosis is a Epidermoid cyst - most commonly called a sebaceous cyst.

They are benign cysts that are filled with keratin and lipid rich debris (Lam, 2020).

2. There are 3 genetic conditions that are associated with developing epidermoid cysts:

- Gardner Syndrome
- Pachyonychia Congenita Type 2
- Basal cell naevus syndrome.

Gardner Syndrome is an inherited disease that is a variant of familial adenomatous polyposis which can present with extra-colonic features including epidermoid cysts. It is important to diagnosis this condition promptly, given the risks of malignant progression from colonic polyps. Other cutaneous features of Gardner syndrome include lipomas, fibromas, leiomyomas and neurofibromas.



Gardner syndrome

Multiple facial epidermal cysts in Gardner syndrome



Reusing this image

Figure 1 - Gardner Syndrome - Derm Net NZ

Question for the group:

1. What is the genetic cause of Gardner Syndrome and how is it inherited?
2. If you saw a patient with multiple epidermoid cysts and suspected Gardner Syndrome, what would your approach be with arranging investigations to confirm a diagnosis?

References:

1. Lam et al (2020) Epidermoid Cysts - DermNet NZ. Available at: <https://dermnetnz.org/topics/epidermoid-cyst> (Accessed 7/11/23)
2. Elyoussfi (2004) Gardner Syndrome - DermNet NZ. Available at: <https://dermnetnz.org/topics/gardner-syndrome> (Accessed 7/11/23)

197 words

Reply



Re: Scenario 3

by [Pooran Outar](#) - Wednesday, 8 November 2023, 2:16 AM

Hi Sophie,

I also agree that an image would add sope to the diagnosis of this case. Nonetheless, I am in accordance with your diagnosis. With respect to your questions, permit me to add a brief discusion on Gardner Syndrome. I would first like to state I have no clinical experience diagnosing or managing this disease but, according to Schwartz, 2023, Gardner syndrome is a genetic condition closely related to Familial Adenomatous Polyposis. The inheritance pattern of Gardner syndrome is autosomal dominant. This means that if a person has a mutation in one copy of the APC gene, they have a 50% chance of inheriting the mutation. This syndrome is characterized by two main features: firstly, the development of numerous growths or polyps in the digestive system, particularly in the colon and rectum, which have a risk of becoming cancerous. Secondly, individuals with Gardner syndrome also experience non-cancerous growths on the skin and in soft tissues, including epidermoid cysts, small skin lumps, and desmoid tumors, which are growths in the connective tissue. When suspecting Gardner syndrome a detailed family and personal medical history should complement physical examination. Endoscopy of the lower GI tract is

necessary to evaluate polyps and Gene testing to detect mutation in APC gene is the only confirmatory test. If positive, endoscopic screenings for polyps can begin as early as age 10 (Schwartz, 2023).

Reference:

Schwartz, R.A. (2023). Dermatologic Manifestations of Gardner Syndrome: Practice Essentials, Pathophysiology, Etiology. eMedicine. [online] Available at: <https://emedicine.medscape.com/article/1093486-overview#a5> [Accessed 8 Nov. 2023].

249 words

Reply

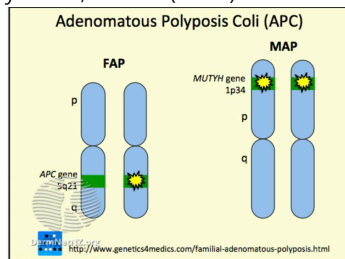


Re: Scenario 3

by [Sadaf Basharat](#) - Thursday, 9 November 2023, 9:44 PM

Hi Sophie, i would like to answer your first question as follow.

Gardner Syndrome is a result of mutations on the adenomatous polyposis coli (APC) gene on chromosome 5q22. The gene plays a role in tumour suppression. Gardner syndrome is inherited as an autosomal dominant, so an affected person has a 50% chance of passing on the gene to each of their children. Elyoussfi, S. et al.(2004).



68 words

Reply



Re: Scenario 3

by [Derek Kayanja \(Tutor\)](#) - Friday, 10 November 2023, 9:38 AM

Thank you Sophie, good post and the correct diagnosis (I didn't include an image as it would have been a give away).

22 words

Reply



Re: Scenario 3

by [Zarmeena Khan](#) - Tuesday, 14 November 2023, 10:35 AM

Hi Sophie, I agree with the diagnosis of Epidermoid cyst, and I would like to answer the second part of your question. Diagnosing Gardner syndrome can involve a combination of clinical evaluation, imaging studies, and genetic testing. If a patient has multiple epidermoid cysts and suspected Gardner syndrome, the following investigations can help with the diagnosis:

1. Clinical Evaluation:

- A thorough medical history should be taken, with attention to gastrointestinal symptoms, family history of colorectal polyps or cancer, and extra-colonic manifestations associated with Gardner syndrome.
- Physical examination should focus on dermatological signs (such as epidermoid cysts), dental abnormalities, and other potential features of Gardner syndrome.

2. Colonoscopy:

- A colonoscopy is a crucial tool for diagnosis because Gardner syndrome is often associated with multiple colorectal polyps. It allows for the visualization and biopsy of polyps to assess their nature and potential for malignancy.

- Screening colonoscopy is recommended at an earlier age for individuals with a family history of Gardner syndrome.

3. Genetic Testing:

- Genetic testing for mutations in the adenomatous polyposis coli (APC) gene is a definitive method for confirming the diagnosis of Gardner syndrome. Blood or saliva samples can be analyzed for APC gene mutations.

- It's particularly important to consider genetic testing for individuals with a family history of Gardner syndrome or familial adenomatous polyposis (FAP).

4. Imaging Studies:

- Radiological imaging, such as abdominal imaging (CT or MRI), may be performed to assess the presence of extra-colonic manifestations, such as desmoid tumors.

5. Dental and Skeletal Evaluation:

- Dental abnormalities and osteomas are common features of Gardner syndrome. Evaluating for these abnormalities can be done through dental panoramic radiographs.

References:

1. Groden, J., Thliveris, A., Samowitz, W., Carlson, M., Gelbert, L., Albertsen, H., ... & Joslyn, G. (1991). *Identification and characterization of the familial adenomatous polyposis coli gene*. Cell, 66(3), 589-600. DOI: [10.1016/0092-8674(91)90111-H]([https://doi.org/10.1016/0092-8674\(91\)90111-H](https://doi.org/10.1016/0092-8674(91)90111-H))

2. Nieuwenhuis, M. H., & Vasen, H. F. (2007). *Correlations between mutation site in APC and phenotype of familial adenomatous polyposis (FAP): a review of the literature*. Critical Reviews in Oncology/Hematology, 61(2), 153-161. DOI: [10.1016/j.critrevonc.2006.07.004](<https://doi.org/10.1016/j.critrevonc.2006.07.004>)

345 words

Reply



Re: Scenario 3

by [Siddharth Saluja](#) - Wednesday, 15 November 2023, 11:49 AM

1. Familial adenomatous polyposis has a phenotypic variation known as Gardner syndrome. Numerous adenomatous polyps along the intestinal mucosal surface with a high risk of malignancy characterize this autosomal dominant illness. In 1951, Gardner gave the syndrome's first description. He reported having malignancies outside the colon in addition to several polyps inside the colon. It is possible for intestinal polyposis, desmoids, osteomas, and epidermoid cysts to be extracolonic presentations. Patients with Gardner syndrome typically manifest with fibromatosis, epidermal cysts, and osteomas of the jaw and skull. Although pruritus, inflammation, and rupture are possible presentations, these signs are frequently found to be asymptomatic. This illness can also present with a variety of non-cutaneous symptoms. (1)

2. Doctors may use genetic testing to find any APC gene abnormalities in patients exhibiting Gardner syndrome symptoms. Colon screening is available to patients whose family members have such mutations at age 10, therefore it is likely that this testing will be done early if the patient has a parent with the problem.

Gardner syndrome is diagnosed based on the existence of (1,2)

More than 100 polyps in the colon and/or rectal area, or less than 100 polyps among patients who have a family history of illnesses associated to FAP

Additional cancers, such as epidermoid cysts, fibromas, and desmoid tumors
connected cancers

A diagnosis comprises:

physical assessment

Colonoscopy X-ray of the mandible and/or the long bones (phalanges, tibia, fibula, femur, and metatarsals).(2)

References

1)Cancer Treatment Centers of America. (2021). Gardner Syndrome: Symptoms, Causes and Treatments. [online] Available at: <https://www.cancercenter.com/risk-factors/gardner-syndrome>.

2.Healthline. (n.d.). Gardner's Syndrome: Symptoms, Diagnosis, and Treatment. [online] Available at: <https://www.healthline.com/health/gardners-syndrome>.

262 words

Reply



Re: Scenario 3

by [Arwaf Benrawwaf](#) - Monday, 11 December 2023, 12:37 PM

Hello Sophie, thank you for your post

Gardner Syndrome is caused by a mutation in the APC (adenomatous polyposis coli) gene, which is located on chromosome 5q21. This gene is responsible for producing a protein that helps regulate cell growth and division. Inheritance of Gardner Syndrome is autosomal dominant, meaning that a person only needs to inherit one copy of the mutated gene from one parent to develop the condition.

If a patient presents with multiple epidermoid cysts and Gardner Syndrome is suspected, further investigations should be arranged to confirm the diagnosis. The following investigations may be helpful:

- Colonoscopy: This is a procedure that allows a doctor to examine the inside of the colon and rectum for polyps or other abnormalities. People with Gardner Syndrome are at an increased risk of developing colon polyps, which can lead to colon cancer if left untreated.
- Genetic testing: This involves analyzing a person's DNA to look for mutations in the APC gene. Genetic testing can confirm a diagnosis of Gardner Syndrome and also identify family members who may be at risk of developing the condition.
- Imaging studies: X-rays, CT scans, or MRI scans may be used to look for bone abnormalities, such as osteomas, which are common in people with Gardner Syndrome.
- Skin biopsy: If a patient has skin lesions or fibromas, a biopsy may be performed to confirm the diagnosis of Gardner Syndrome.

It is important to note that not all people with Gardner Syndrome will have all of these features, and some people may have only mild symptoms. Therefore, a thorough medical history and physical examination are also important in making a diagnosis.

References:

- [rarediseases.org](https://rarediseases.org/gard-rare-disease/gardner-syndrome/). (2022). *GARD Rare Disease Information - Gardner syndrome - National Organization for Rare Disorders*. [online] Available at: <https://rarediseases.org/gard-rare-disease/gardner-syndrome/> [Accessed 11 Dec. 2023].

- [www.cancer.org](https://www.cancer.org/cancer/types/colon-rectal-cancer/detection-diagnosis-staging/how-diagnosed.html). (2023). *Testing for Colorectal Cancer | How Is Colorectal Cancer Diagnosed?* [online] Available at: <https://www.cancer.org/cancer/types/colon-rectal-cancer/detection-diagnosis-staging/how-diagnosed.html>.

- National Cancer Institute (2019). *Genetics of Colorectal Cancer*. [online] National Cancer Institute. Available at: <https://www.cancer.gov/types/colorectal/hp/colorectal-genetics-pdq>.

334 words

Reply

**Re: Scenario 3**by [Pooran Outar](#) - Wednesday, 8 November 2023, 1:44 AM**(1) Epidermoid cyst**

(2) The following conditions maybe associated with epidermoid cyst, according to Fromm, 2022: "Congenital epidermoid cyst, Acne, Trauma or surgery, HPV infection, UV exposure, bening or malignant conditions occuring near pilosebaceous unit and hereditary conditons like Gardner syndrome, basal cell nevus syndrome, and pachyonychia congenita" (Fromm, 2022).

(3) The most likely diagnosis is: **Adiposis Dolorosa (Dercum Disease)**

Hansen criteria for diagnosis or Dercum disease inlclude (Munguia et al., 2021):

- Obesity
- Chronic pain > 3 months in fat tissue
- Palpable fatty lipomas



Image 1 - Painful Lipomas (Ryder, 2014)

Differential diagnosis according to medsacpe include (McGevna, 2023):

- Fibromyalgia
- Panniculitis
- Endocrine disorders,
- Multiple symmetric lipomatosis (Madelung syndrome)
- Multiple familial lipomatosis
- Benign adipose tissue tumors.

Therapuetic options include (McGevna, 2023):

- Analgesics
- Liposuction
- Lipectomy
- Manual lymphatic drainage
- Minimal incision technique.

Question:

Some authors suggest that Dercum disease is just an expression of familial multiple lipomas. What evidence is there to support or refute this hypothesis?

References:

- Fromm, L.J. (2022). Epidermal Inclusion Cyst: Background, Pathophysiology, Etiology. *eMedicine*. [online] Available at: <https://emedicine.medscape.com/article/1061582-overview#a7> [Accessed 8 Nov. 2023].
- McGevna, L.F. (2023). Adiposis Dolorosa (Dercum Disease): Practice Essentials, Background, Pathophysiology. *eMedicine*. [online] Available at: <https://emedicine.medscape.com/article/1082083-overview#a5> [Accessed 8 Nov. 2023].
- Munguia, N., Mozayeni, B.R., Wright, T.F. and Herbst, K.L. (2021). Dercum's disease: estimating the prevalence of a rare painful loose connective tissue disease. *Future Rare Diseases*, 1(1), p.FR4. doi:<https://doi.org/10.2217/frd-2020-0004>.
- Ryder, E. (2014). *Dercum disease. Adiposis dolorosa* / *DermNet NZ*. [online] [dermnetnz.org](https://dermnetnz.org/topics/dercum-disease). Available at: <https://dermnetnz.org/topics/dercum-disease>.

238 words

Reply

**Re: Scenario 3**

by [Derek Kayanja \(Tutor\)](#) - Friday, 10 November 2023, 9:39 AM

Thank you Pooran, good post and correct diagnosis

8 words

Reply

**Re: Scenario 3**

by [Oyeombiliyetu Jason](#) - Wednesday, 8 November 2023, 3:03 AM

Q1: Infected Epidermal inclusion cyst, shown by:

1. Male predilection (2:1 ratio).
2. 38y/o, primarily affecting young to middle-aged people (20-40 year olds).
3. Presents as a well-demarcated, cystic lesion within the dermis.
4. Characterized by a central punctum or opening in the lesion.
5. When infected, the cyst becomes red and emits an odor.

Q2: Associated conditions:

1. Gardner syndrome
2. Pachyonychia congenita type 2
3. Basal cell nevus syndrome
4. Favre-Racouchot syndrome
5. Presence of comedones (blackheads and whiteheads)
6. Medication-related associations, such as the use of BRAF inhibitors (e.g., vemurafenib), imiquimod, and cyclosporin.

Q3: Dercum's disease, due to:

1. Increased BMI (Body Mass Index) of 30kg/m²
2. Co-occurring psychiatric conditions, including depression.
3. Manifestation of multiple, painful lipomas.

References

1. B Weir et al. (2023). 'Epidermal inclusion cyst' National Library of Medicine. Available at: <https://www.ncbi.nlm.nih.gov/books/NBK532310/>. Accessed: 08 November 2023.
2. E Kucharz et al.(2019). 'Dercum's disease (adiposis dolorosa): a review of a clinical presentation and management' Reumatologia. 57(5): pg 281-287. Available at: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6911249/>. Accessed: 08 November 2023.

169 words

Reply

**Re: Scenario 3**

by [Derek Kayanja \(Tutor\)](#) - Friday, 10 November 2023, 9:42 AM

Thank you Oyeombiliyetu, good post.

5 words

Reply

**Re: Scenario 3**

by [Bhavika Patel](#) - Sunday, 12 November 2023, 10:04 PM

Thank you Oyeombiliyetu, I wanted to add more information about the clinical features of Dercum's disease which I found in my research.

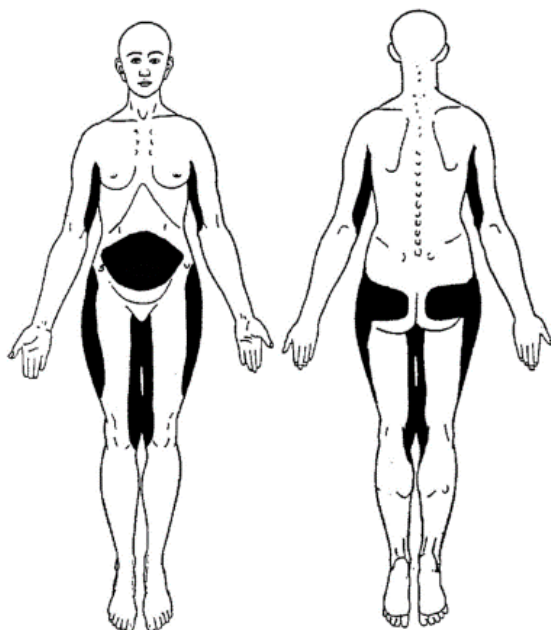
Dercum's disease (Ander's syndrome, Adipose Dolorosa, Dercum-Vitaut Syndrome)

Initially established diagnosis in 1892 Francis Xavier Dercum who noted multiple "painful subcutaneous lipomas" across the trunk and extremities, predominantly in postmenopausal women. There are 3 variants (Priya, 2018):

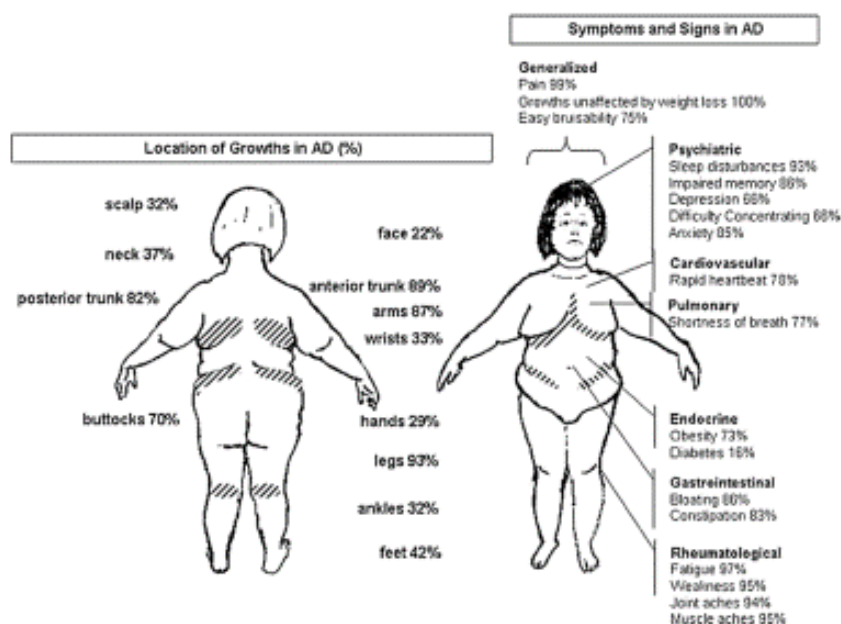
Type 1: generalised diffuse form with painful lipomas affecting the hips and knees.

Type 2: more generalised nodular form, widespread painful lipomas affecting "fat bearing area" of the abdomen, fat folds and intertrigo regions directly correlating to generalised obesity.

Type 3: multiple intensely painful lipomas in a more localised nodular form.



The figure above demonstrates the locations of generalised lipomas in Dercum's disease and figure below shows the associating signs and symptoms (Hannsson et al, 2012). It is thought to be an autosomal dominant disorder particularly affecting MEN1 gene or A8344 of mitochondrial DNA (Hannsson et al, 2012).



Priya, S. S. (2018) Dercum's Disease (Adiposis Dolorosa). *International Journal of Nursing Education and Research*. [Online] 6 (4). (Accessed 12th November 2023)

Hansson, E., Svensson, H., & Brorson, H. (2012) Review of Dercum's disease and proposal of diagnostic criteria, diagnostic methods, classification and management. *Orphanet journal of rare diseases*. [Online] 7 (1), 23–23. (Accessed 12th November 2023)

215 words

Reply



Re: Scenario 3

by [Siddharth Saluja](#) - Friday, 17 November 2023, 10:37 AM

Hello Bhavika,

From your experience in clinical practise, clinically would you use a dermatoscope to work out the extent and progression of the disease or is there another way of quantifying the extent off the spread, I assume a lot of it would be histopathological diagnosis for the final confirmation of the diagnosis?

53 words

Reply



Re: Scenario 3

by [Jillian Simpson](#) - Wednesday, 8 November 2023, 11:39 AM

1. Given the age, sex, and description of a well-demarcated nodule with a punctum that releases a smelly discharge I think this is a epidermoid cyst (Lam et al, 2020).

2. There are a few conditions that are associated with developing multiple epidermoid cysts and include:

a) Gardner's syndrome, this is a rare autosomal dominant disorder. It is associated with multiple epidermoid cysts (usually irregularly distributed on the face, scalp and extremities), polyposis of the colon, osteomas (mainly of the skull) and fibromas (PCDS, 2022)

b) Pachyonychia congenita type 2, this is a rare, inherited disorder of keratinisation. Pachyonychia congenita is characterised by thickened skin of the palms and soles, thickened nails, and white patches in mucous membranes. The specific features depend on which keratin gene is affected (Abeyakirithi et al, 2020)

c) Basal cell naevus syndrome also known as Gorlin syndrome - this is a rare genetic cancer syndrome characterised by multiple early onset basal cell carcinomas, odontogenic keratocysts and other tumours, and other abnormalities (Oakley et al, 2021).

3. Given the age, obesity, depression and multiple painful lipomas, I would want to explore dercum disease as a diagnosis. It would be useful to find out if any other signs such as painful subcutaneous plaques or ecchymoses (bruises) without noticed trauma, where also present (Ryder et al, 2014).

References:

Lam, M. Michael, G, Oakley, A, Mitchell, G. Epidermoid cyst | DermNet NZ. [online] Available at:

<https://dermnetnz.org/topics/epidermoid-cyst>. [Accessed 8/11/23]

PCDS. (2022). Epidermoid cyst (syn. epithelial cyst. Sebaceous cyst is a misnomer). [online] Primary Care Dermatology Society. Available at: <https://www.pcds.org.uk/clinical-guidance/epidermoid-cyst>. [Accessed 8/11/23]

Abeyakirithi, S. Chang, F.C. Oakley, A. (2020) Pachyonychia congenita | DermNet NZ. [online] Available at:

<https://dermnetnz.org/topics/pachyonychia-congenita>. [Accessed 8/11/23]

Oakley, A. Grossman, M. Keefe, M. Mitchell, G. (2021). Basal cell naevus syndrome (BCNS) | DermNet NZ.

[online] Available at: <https://dermnetnz.org/topics/basal-cell-naevus-syndrome>. [Accessed 8/11/23]

Ryder, E. Oakley, A. Mitchel, G. (2014). Dercum disease. Adiposis dolorosa | DermNet NZ. [online] Available at:

<https://dermnetnz.org/topics/dercum-disease>. [Accessed 8/11/23]

326 words

Reply



Re: Scenario 3

by [Derek Kayanja \(Tutor\)](#) - Friday, 10 November 2023, 9:42 AM

Thank you Jillian, good post.

5 words

Reply



Re: Scenario 3

by [Xiao Liey Ding](#) - Wednesday, 8 November 2023, 3:25 PM

Question 1:

Sebaceous cyst (epidermoid cyst)

Supporting points: patient's age (adult), sex (male), characteristic of punctum and cystic lump content

Question 2:

Epidermoid cyst can occur secondary to **ruptured follicle linked to acne or if the gland does not grow properly or post injury.**

Some **hereditary conditions** have been associated with epidermoid cysts such as Gardner syndrome and Gorlin syndrome (basal cell nevus syndrome) (Patrick M. Zito and Richard Scharf, 2023). Chronic sun damage can also cause epidermoid cyst in elderly with Favre-Racouchot syndrome. Pachyonychia congenita type 2 is also linked to epidermoid cyst (Megan L. et al, 2020).

Jillian has posted the symptoms and signs of Gardner syndrome, Gorlin syndrome and Pachyonychia congenita type 2 in her post. Favre-Racouchot syndrome consists of open comedones, closed comedones, and epidermal cysts located on the temples, lateral cheeks, periorbital areas, and nose of facial skin that has been sun-damaged.

Question 3:

Dercum disease, characterized by multiple painful lipomas.

References:

Patrick M. Zito and Richard Scharf (2023). Epidermoid Cyst. [online] National Library of Medicine. Available at: <https://www.ncbi.nlm.nih.gov/books/NBK499974/> (Accessed on 8 Nov 2023)

Megan Lam, Michael G, Oakley A. (2020) Epidermoid Cysts. [online] DermNet NZ. Available at: <https://dermnetnz.org/topics/epidermoid-cyst> (Accessed on 8 Nov 2023)

199 words

Reply

**Re: Scenario 3**by [Derek Kayanja \(Tutor\)](#) - Friday, 10 November 2023, 9:43 AM

Thank you Xiao Liey, good post.

6 words

Reply

**Re: Scenario 3**by [Adam Tollitt](#) - Wednesday, 8 November 2023, 8:21 PM

Hi Pooran, Oyeombiliyetu, Jillian and Ding,

Thank you all for your explanations of Dercum Disease.

This is not a condition that I have previously heard of until reading your comments. As already mentioned, it is a relatively rare disease and prevalence data is limited. I have found one study that estimated prevalence in a USA population of 0.65% (Munguia et al., 2021).

Although a rare condition, I wonder how many patients with Dercum Disease are undiagnosed or misdiagnosed? Many of the core symptoms/features appear to overlap with other more commonly diagnosed chronic pain conditions and/or fibromyalgia.

References

Munguia, N., Mozayeni, B.R., Wright, T.F. and Herbst, K.L. (2021). Dercum's disease: estimating the prevalence of a rare painful loose connective tissue disease. *Future Rare Diseases*, 1(1), p.FRD4.
doi:<https://doi.org/10.2217/frd-2020-0004>.

126 words

Reply

**Re: Scenario 3**by [Derek Kayanja \(Tutor\)](#) - Friday, 10 November 2023, 9:43 AM

Adam, thank you for your contribution to the discussion.

9 words

Reply

**Re: Scenario 3**by [Derek Kayanja \(Tutor\)](#) - Friday, 10 November 2023, 9:47 AM

Hi Group,

Thank you all who have posted a response to scenario three, you have all arrived at the correct diagnosis for both patients.

Our 38 year old male patient presented with an Epidermoid cyst, and the associated condition is Gardner syndrome. Our 56-year-old female patient with a BMI of 30 kg/m² and depression presented with a diagnosis of Dercum's disease

Epidermoid cysts, also known as "sebaceous cysts", are the most common cutaneous cyst. They are most common on the face and upper trunk, but can occur anywhere. They present as an asymptomatic, well-demarcated, skin-coloured to yellowish dermal nodule which may grow to several centimetres, and may have

a punctum (represents the follicle from which the cyst derived). They are derived from the follicular infundibulum due to follicular disruption. The cyst contents can be expressed with pressure and may have an odour. Rupture of the cyst wall can cause a painful inflammatory reaction. Risk factors include acne vulgaris, Gardner syndrome, naevoid basal cell carcinoma syndrome. Rarely BCC/SCC develops within the cyst. Treatment is by excision of entire cyst and cyst wall, as if the cyst wall is not removed, it will recur.

Dermoid cysts are discrete subcutaneous 1-4 cm nodules, most commonly seen in infants along the embryonic fusion plane, especially around the eyes. Histopathology shows a cyst lined by stratified squamous epithelium including a granular layer and they contain hair, sebaceous lobules, eccrine glands, apocrine glands or smooth muscle. Treatment is by excision, however, MRI/USS (ultrasound scan) may be necessary first, to exclude connection to CNS, as neuronal heterotopias are a differential diagnosis.

Lipoma is a common benign tumour of mature fat. It is an asymptomatic, soft, rubbery, mobile subcutaneous nodule with normal overlying epidermis, and can arise at any site, although it is more common on the neck, proximal limbs, forearms and buttocks. Lipomas are slightly more common in men and those over 30 years old. They are usually solitary, however, 5-10% of people will have multiple. Risk factors include: obesity, diabetes and hypercholesterolaemia. Multiple lipomas are seen in familial multiple lipomatosis, Madelung's disease, adiposis dolorosa (Dercum's disease), Gardner syndrome, Bannayan-Riley-Ruvalcaba syndrome, proteus syndrome and tuberous sclerosis. Treatment involves simple excision if they are symptomatic, otherwise they can be left alone.

Milium is a common tiny superficial epidermoid cyst, 1-2 mm white to yellow subepidermal papule. Milia are most common on the face. They occur in 40-50% of infants, usually resolving spontaneously by 4 weeks of age. They can be primary or secondary; secondary following blistering processes, superficial ulceration from trauma, cosmetic procedures or at sites of topical corticosteroid-induced atrophy. Associated syndromes include: Marie-Unna Hypotrichosis (hereditary trichodysplasia), BCC-associated syndromes, Rombo syndrome and Bazex syndrome. Treatment is to incise the epidermis overlying the milium with a needle and express the milium. Laser ablation, electrodesiccation or topical retinoids can sometimes be helpful in multiple milia.

Vellus hair cysts are numerous, asymptomatic, small, dome-shaped, skin-coloured or darkly pigmented papules, most common on the trunk. They may be inherited in an autosomal dominant pattern. Multiple lesions are termed "eruptive vellus hair cysts". They can be associated with steatocystoma multiplex and pachyonychia congenita. They do not resolve spontaneously and can be treated by incision and drainage, needle evacuation, topical retinoids, topical lactic acid or laser ablation.

Gardner syndrome consists of polyposis of the colon, eventually leading to colon carcinoma if not treated, odontomas, multiple epidermoid cysts, osteomas, leiomyomas, desmoids fibromatosis and congenital hypertrophy of the retinal pigment epithelium. It is caused by a mutation in the APC gene.

Dercum's disease (adiposis dolorosa) usually presents in postmenopausal females with multiple painful lipomas on the arms, trunk and periarticular tissue. Pain may be intermittent and is usually caused by the lipoma compressing adjacent nerves. It is associated with weakness and psychiatric symptoms.

628 words

Reply



Re: Scenario 3

by [Ayten Sadek](#) - Friday, 10 November 2023, 6:39 PM

Question 1:

The most likely diagnosis is Epidermoid cyst (Sebaceous cyst)

They are usually skin-colored or yellowish in appearance. They may have a punctum through which a thick yellow foul smelling material can be expressed. This material is a mixture of dead skin cells and oils.

Question 2:

Associated conditions:

1- Gardner syndrome: a rare genetic disorder and is a subtype of familial adenomatous polyposis (FAP) characterized by the development of numerous polyps in the colon and rectum. It is associated with extra-colonic features, which can include: Multiple Epidermoid Cysts, Osteomas, Dental Abnormalities and Desmoid tumors.

2- Pachyonychia congenita type 2: Presentation depends on which keratin gene affected but generally characterized by thickened nails (pachyonychia), Thickened palms and soles, may also affect the mouth causing leukokeratosis

3- Basal cell naevus syndrome: is a rare genetic cancer syndrome characterised by multiple basal cell carcinomas, palmer and planter pits, skeletal abnormalities, calcification of falx cerebri and skeletal abnormalities.

Question 3:

Dercum disease due to increased BMI and painful lipomas.

.References:

Dermnetnz.org(2020) Epidermoid cyst(online) available at: <https://dermnetnz.org/topics/epidermoid-cyst> (accessed on 9/11/2023)

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202 words

Reply



Re: Scenario 3

by [Derek Kayanja \(Tutor\)](#) - Sunday, 12 November 2023, 2:30 PM

Ayten, thank you for your interesting post.

7 words

Reply



Re: Scenario 3

by [Rahima Abdulaziz Ahmed](#) - Saturday, 11 November 2023, 2:39 PM

Thank you all colleagues and Tutor for the wonderful discussions.

Q3 ;

Dercum disease is associated with conditions like;

Endocrine Disorders; eg hormonal imbalances , myxedema

Autoimmune disorders ;eg rheumatoid arthritis , lupus

Nervous system abnormalities; eg generalized weakness, depression, impaired memory , sleep disturbances and irritability.

References;

Hansson .E , Svensson. H and Brorson .H (2012) *Review of Dercum disease* (online) Available at <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3444313/> (.Accessed on 11/11/23)

Herbst. K (2020) *Dercum's Disease* (online) Available at <https://rarediseases.org/rare-diseases/dercums-disease/> (Accessed on 11/11/23)

79 words

Reply

**Re: Scenario 3**by [Derek Kayanja \(Tutor\)](#) - Sunday, 12 November 2023, 2:30 PM

Rahima, thank you for your post.

6 words

[Reply](#)**Re: Scenario 3**by [Arwaf Benrawwaf](#) - Sunday, 12 November 2023, 7:21 PM

Q1. The most likely diagnosis for the 38-year-old man's skin nodule is an epidermoid cyst or sebaceous cyst.



(dermnetnz.org,2020)

Q2. These types of skin lumps are associated with conditions such as: (dermnetnz.org,2020)

- acne
- Lipoma
- Trichilemmal cyst
- Myxoid pseudocyst
- Dermoid cyst
- folliculitis
- hidradenitis suppurativa.
- Human Papilloma virus (HPV)

Q3. The 56-year-old woman with multiple painful lipomas and a BMI of 30 kg/m² is most likely to be associated with Dercum disease, also known as adiposis dolorosa. This condition is characterized by the presence of painful lipomas in the subcutaneous tissue, often in association with obesity and depression. (Hansson, E.,2012)

References:

- James, W.D., Berger, T. and Elston, D. (2015). *Andrews' Diseases of the Skin : Clinical Dermatology*. London: Elsevier Health Sciences.

- Hansson, E., Svensson, H. and Brorson, H. (2012). Review of Dercum's disease and proposal of diagnostic criteria, diagnostic methods, classification and management. *Orphanet Journal of Rare Diseases*, 7(1), p.23. doi:<https://doi.org/10.1186/1750-1172-7-23>.

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176 words

Reply



Re: Scenario 3

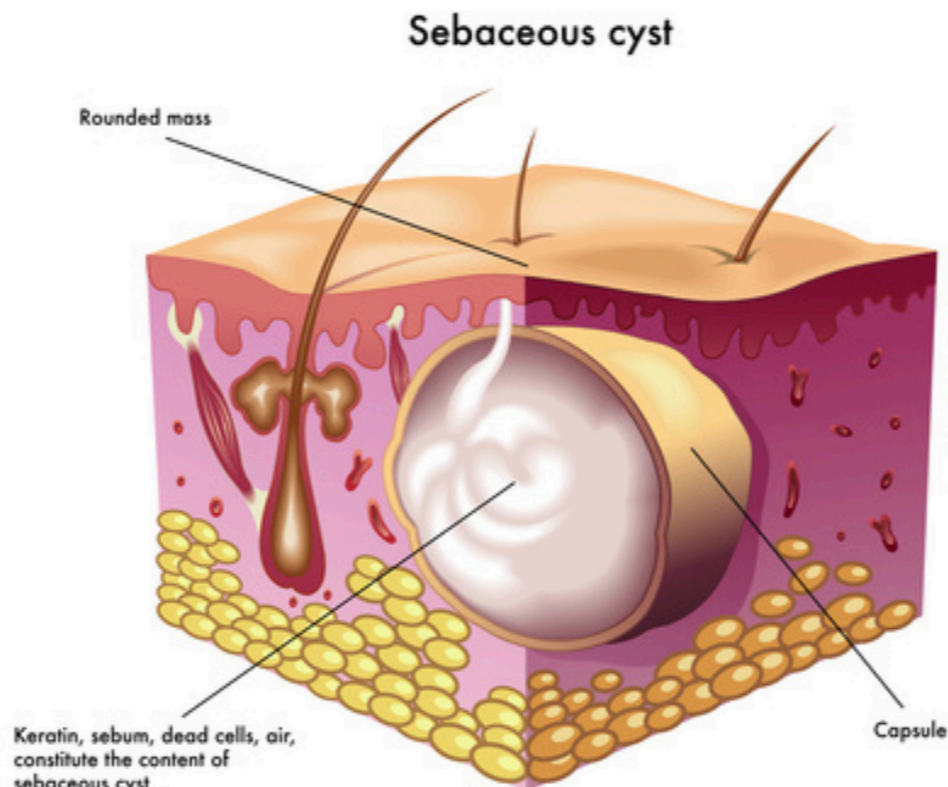
by [Zarmeena Khan](#) - Tuesday, 14 November 2023, 11:08 AM

Hi everyone, sorry to join this chat late as I was on call,

excellent discussion by everyone,

I agree with the diagnosis of Epidermoid cysts, also known as epidermal inclusion cysts or sebaceous cysts, which are common non-cancerous cysts that develop beneath the skin. These cysts form when the top layer of skin cells (epidermis) proliferates within a confined space, creating a closed sac filled with keratin, a protein found in skin, hair, and nails. The cysts are typically slow-growing and may vary in size.

The precise cause of epidermoid cysts is often related to trauma or blockage of the hair follicles, leading to the entrapment of skin cells. They can occur on any part of the body but are most commonly found on the face, neck, and trunk.



(The Dermatology Specialists, 2021)

Conditions associated with epidermoid cysts include:

1. Gardner Syndrome:

- As previously mentioned, Gardner syndrome, a subtype of familial adenomatous polyposis (FAP), is associated with the development of epidermoid cysts along with multiple colorectal polyps and other extra-colonic manifestations.

(Habif, T. P. 2016)

2. I would like to add - Muir-Torre Syndrome:

- This is a variant of Lynch syndrome (hereditary non-polyposis colorectal cancer) characterized by the presence of sebaceous adenomas, sebaceous carcinomas, and keratoacanthomas. Epidermoid cysts may also be observed in individuals with Muir-Torre syndrome.

3. Gorlin Syndrome (Nevoid Basal Cell Carcinoma Syndrome):

- Individuals with Gorlin syndrome may develop multiple basal cell carcinomas. While epidermoid cysts are not a primary feature, they have been reported in association with this syndrome.

4. Inclusion Cyst Syndrome:

- Inclusion cyst syndrome is a rare genetic disorder characterized by the presence of multiple epidermoid cysts.

It's important to note that while these conditions may be associated with epidermoid cysts, the presence of a single epidermoid cyst is more commonly unrelated to these syndromes and is often a benign, sporadic occurrence.

C. I agree regarding the diagnosis of Dercum's disease,

This a rare disorder characterized by the development of painful, fatty deposits or lipomas beneath the skin. These lipomas are typically found on the trunk, upper arms, and thighs. The exact cause of Dercum's disease is not well understood, and it predominantly affects middle-aged women.

The main symptoms of Dercum's disease include the painful growth of fatty tissue, generalized weakness, and fatigue. The pain associated with this condition can be chronic and debilitating, affecting the individual's quality of life.

Diagnosis is based on clinical evaluation, and there are no specific laboratory tests or imaging studies that can definitively confirm Dercum's disease. Treatment is generally focused on managing symptoms and may include pain management, physical therapy, and, in some cases, surgical removal of painful lipomas.

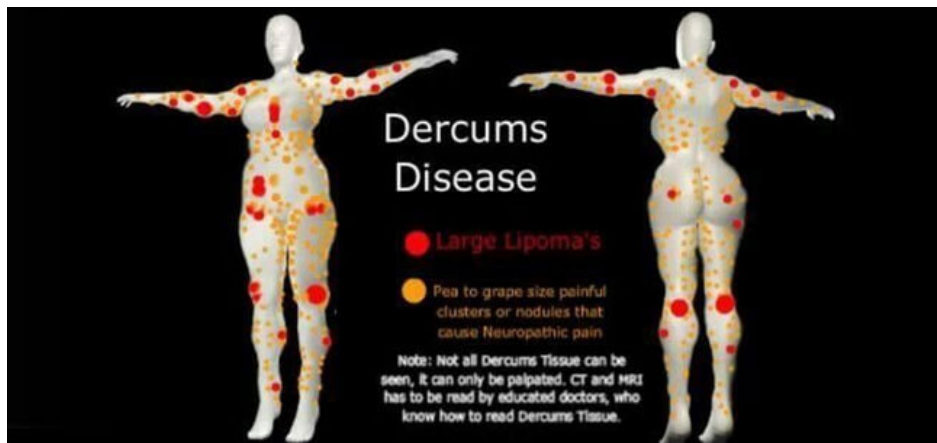
(Dermnetnz.org, 2022)

Types of Dercums

TYPE 1: Painful nodular fat around the joints.

TYPE 2: Painful nodular fat anywhere from head to toe (whole body). Often accompanied by generalized obesity.

TYPE 3: Larger lipomas that may have a capsule and attached connective tissue; often in the absence of obesity. Most men have Type 3



(Lymphatic Therapy Services, 2020)

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5. Lymphatic Therapy Services. (2020). *Dercums Disease Treatment San Diego - Lymphatic Therapy Services*. [online] Available at: <https://lymphatictherapyservices.com/dercums-san-diego/> [Accessed 14 Nov. 2023].

635 words

Reply

◀ Scenario 2

◀ My Reflections

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